

PYTHON FUNDAMENTALS ASSIGNMENT 1

STRING AND TUPLE

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STRINGS:

1. String Concatenation:

Write a Python program that takes two strings i.e string 1 "Hello ", string 2 get name as input from the user and concatenates them together. Display the concatenated string as the output.

Now concatenate string 3 ", welcome to Python programming" to the existing string and display the output string.

```
[1] ✓ 12s ▶ string1 = "Hello"
string2= input("Enter your name")
... Enter your nameAlice
```

```
[4] ✓ 0s ▶ result = string1 + string2
print(result)
... Hello Alice
```

```
[5] ✓ 0s ▶ string3= ", welcome to python programming."
result = result + string3
print(result)
... Hello Alice, welcome to python programming.
```

2. String Slicing and Indexing:

Write a Python program using the above concatenated string as input and performs the following tasks:

- Print the first character of the string.

```
[7]
✓ 0s
▶ print("First character : ", result[0])
... First character : H
```

- b. Print the last character of the string.

```
[8]
✓ 0s
▶ print("Last character : ", result[-1])
... Last character : .
```

- c. Print the first 5 characters of the string.

```
[9]
✓ 0s
▶ print("First 5 characters : ", result[:5])
... First 5 characters : Hello
```

- d. Print the last 11 characters of the string.

```
[10]
✓ 0s
▶ print("Last 11 characters : ", result[-11:])
... Last 11 characters : rogramming.
```

- e. Print the string in reverse.

```
[12]
✓ 0s
▶ print("Reverse : ", result[::-1])
... Reverse : .gnimmargorp nohtyp ot emoclew ,ecila olleH
```

- f. Use slicing and print the word "Python" from the existing string.

```
[13]
✓ 0s
▶ print("Word python : ", result[result.index("python"):result.index("python")+6])
... Word python : python
```

3. String Methods:

Write a Python program that takes a string, strM = "Python beginner tutorial" and perform the following tasks:

```
[14] ✓ 0s ▶ strM = "Python beginner tutorial"
```

- a. Convert the sentence to uppercase.

```
▶ print("Uppercase : ", strM.upper())  
... Uppercase : PYTHON BEGINNER TUTORIAL
```

- b. Convert the sentence to lowercase.

```
[16] ✓ 0s ▶ print("Lowercase : ", strM.lower())  
... Lowercase : python beginner tutorial
```

- c. Use Capitalize and return the sentence to the original input form.

```
[17] ✓ 0s ▶ print("Capitalized : ", strM.capitalize())  
... Capitalized : Python beginner tutorial
```

- d. Count the total number of occurrences of character 't' in the string.

```
[18] ✓ 0s ▶ print("Number of t : ", strM.count('t'))  
... Number of t : 3
```

- e. Replace all occurrences of "Python" with "Machine Learning" in the input string strM = "Python beginner tutorial"

```
[19] ✓ 0s ▶ print("Replaced : ", strM.replace("Python", "Machine Learning"))  
... Replaced : Machine Learning beginner tutorial
```

Tuples (Creation, Modification and Access) :

Create 1st tuple with values -> (10, 20, 30), 2nd tuple with values -> (40, 50, 60):

```
[20]
✓ 0s t1 = (10, 20, 30)
      t2 = (40, 50, 60)
```

- a. Concatenate the two tuples and store it in "t_combine"

```
[21]
✓ 0s t_combine = t1 + t2
      print("Combined tuple : ", t_combine)
... Combined tuple : (10, 20, 30, 40, 50, 60)
```

- b. Repeat the elements of "t_combine" 3 times

```
[22]
✓ 0s print("Repeated tuple:", t_combine * 3)
... Repeated tuple: (10, 20, 30, 40, 50, 60, 10, 20, 30, 40, 50, 60, 10, 20, 30, 40, 50, 60)
```

- c. Access the 3rd element from "t_combine"

```
[23]
✓ 0s print("3rd element : ", t_combine[2])
... 3rd element : 30
```

- d. Access the first three elements from "t_combine"

```
[24]
✓ 0s print("First three elements:", t_combine[:3])
... First three elements: (10, 20, 30)
```

- e. Access the last three elements from "t_combine"

```
[25]
✓ 0s print("Last three elements:", t_combine[-3:])
... Last three elements: (40, 50, 60)
```